```
clear; clc; close all;
run(fullfile(fileparts(fileparts(mfilename('fullpath'))),
'setup_project.m'));

m = 4;
seeds = 1:30;
names = {'Random', 'Highest degree', 'Closest to g'};
scores = zeros(numel(seeds), 3);

rng(0);
for i = 1:numel(seeds)
    D = make_deployment(seeds(i));
    scores(i, 1) = score_mission(D, pick_random(D, m));
    scores(i, 2) = score_mission(D, pick_degree(D, m));
    scores(i, 3) = score_mission(D, pick_closest(D, m));
end

rowNames = arrayfun(@(s) sprintf('seed_%d', s), seeds, 'UniformOutput',
false);
T = array2table(scores, 'VariableNames', {'Random', 'Degree', 'Closest'}, ...
                'RowNames', rowNames);
disp(T);

fprintf('\n%-16s %8s %8s %8s\n', 'Strategy', 'Mean', 'Median', 'Worst');
for j = 1:3
    fprintf('%-16s %8.2f %8.2f %8.2f\n', names{j}, ...
        mean(scores(:,j)), median(scores(:,j)), min(scores(:,j)));
end

figure; hold on;
for j = 1:3
    plot(j * ones(size(scores,1), 1) + 0.08 * randn(size(scores,1), 1), ...
         scores(:,j), 'o');
    plot([j-0.3 j+0.3], median(scores(:,j)) * [1 1], 'k-', 'LineWidth', 2);
end
hold off;
set(gca, 'XTick', 1:3, 'XTickLabel', names);
ylabel('Score J (%)'); grid on;
title(sprintf('Baseline score distributions over %d practice deployments',
numel(seeds)));

pickers = {@pick_random, @pick_degree, @pick_closest};
for j = 1:3
    [worst, idx] = min(scores(:, j));
    D = make_deployment(seeds(idx));
    S = pickers{j}(D, m);
    fprintf('%s worst: seed %d, J = %.2f\n', names{j}, seeds(idx), worst);

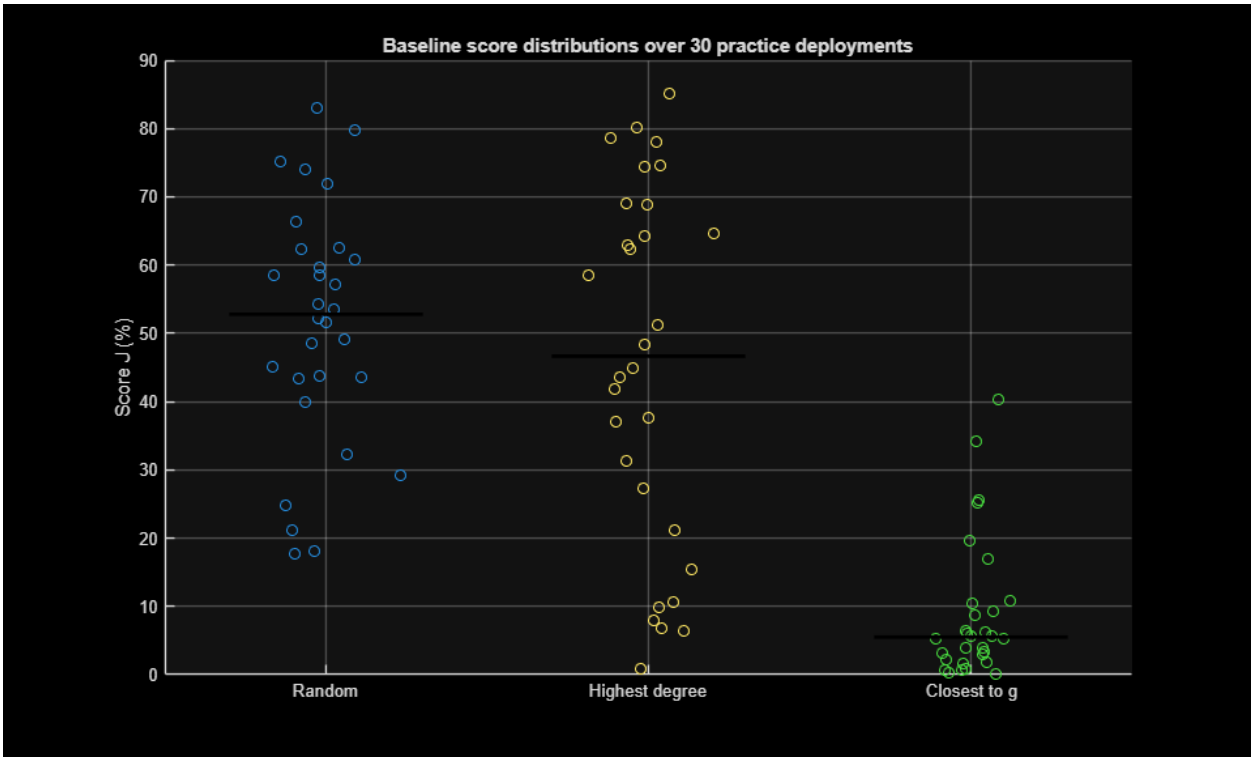
    figure; hold on;
    plot(graph(D.G), 'XData', D.P0(:,1), 'YData', D.P0(:,2), ...
         'NodeColor', [0.5 0.5 0.5], 'EdgeColor', [0.85 0.85 0.95],
'MarkerSize', 3);
```

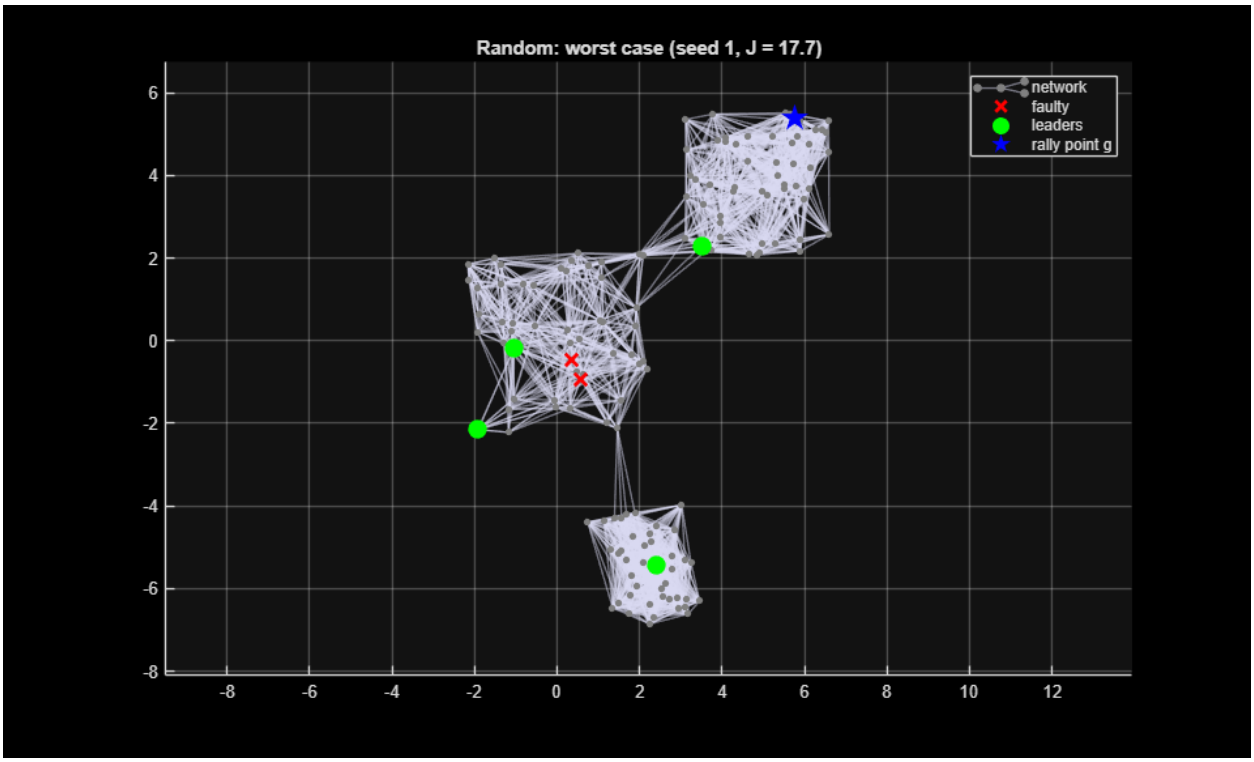
```
    plot(D.P0(D.faulty,1), D.P0(D.faulty,2), 'rx', 'MarkerSize', 10,
'LineWidth', 2);
    plot(D.P0(S,1), D.P0(S,2), 'go', 'MarkerSize', 10, 'MarkerFaceColor',
'g');
    plot(D.g(1), D.g(2), 'bp', 'MarkerSize', 14, 'MarkerFaceColor', 'b');
    axis equal; grid on; hold off;
    legend('network', 'faulty', 'leaders', 'rally point g');
    title(sprintf('%s: worst case (seed %d, J = %.1f)', names{j},
seeds(idx), worst));
end
```

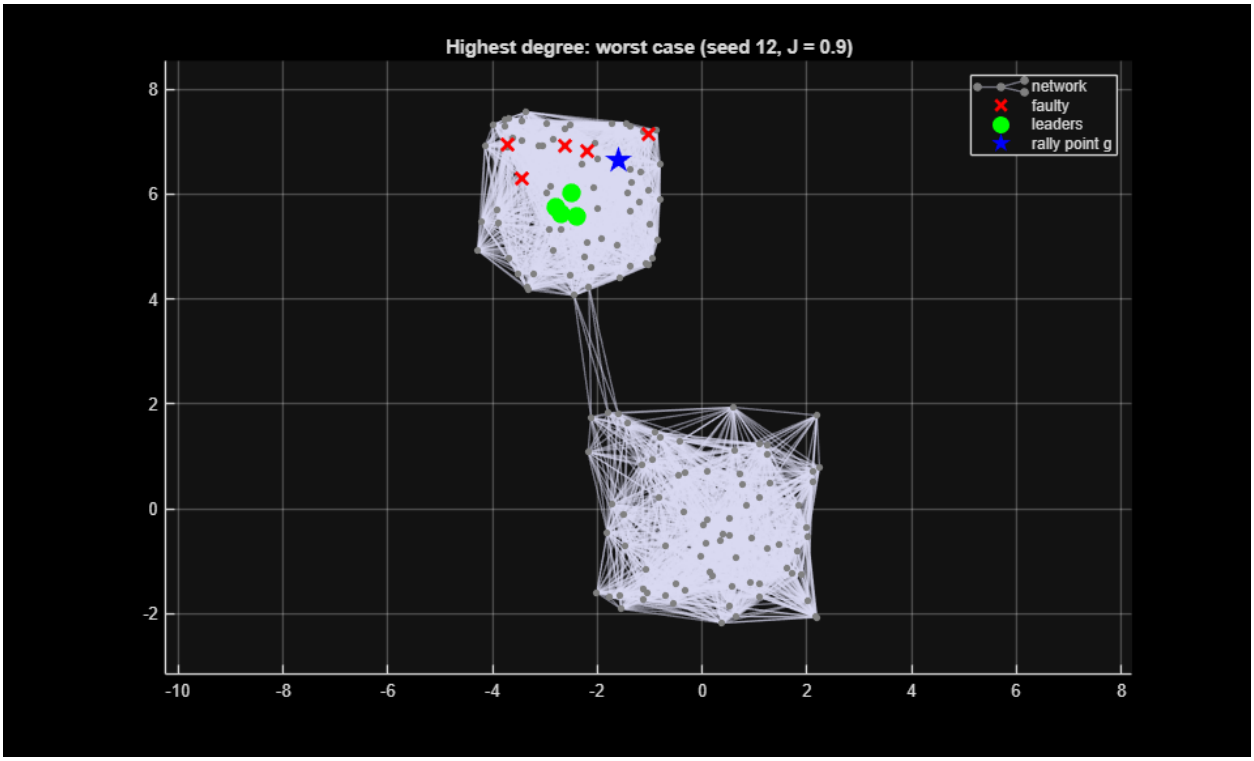
	Random	Degree	Closest
seed_1	17.693	10.705	9.2003
seed_2	29.107	6.8411	5.3592
seed_3	32.234	37.689	2.9157
seed_4	43.652	68.795	0.34165
seed_5	43.305	62.886	0.62493
seed_6	18.055	21.094	6.2696
seed_7	59.625	48.32	40.299
seed_8	79.806	62.414	3.2273
seed_9	58.418	6.4877	5.1796
seed_10	49.163	27.344	16.999
seed_11	45.042	44.991	19.628
seed_12	83.097	0.91875	0.79427
seed_13	73.961	31.215	3.9781
seed_14	24.767	43.507	1.5416
seed_15	62.438	64.643	5.7378
seed_16	53.487	15.486	10.454
seed_17	71.822	78.102	5.6529
seed_18	21.078	37.109	10.898
seed_19	60.814	69.075	3.9716
seed_20	57.119	74.449	6.3433
seed_21	54.369	85.16	1.8404
seed_22	51.512	41.76	34.194
seed_23	43.755	58.523	0.66638
seed_24	75.116	78.575	0.18736
seed_25	52.101	51.286	25.622
seed_26	39.884	9.8202	8.6377
seed_27	62.29	74.52	3.345
seed_28	66.309	64.176	25.112
seed_29	48.472	7.9235	2.1932
seed_30	58.528	80.159	6.0174

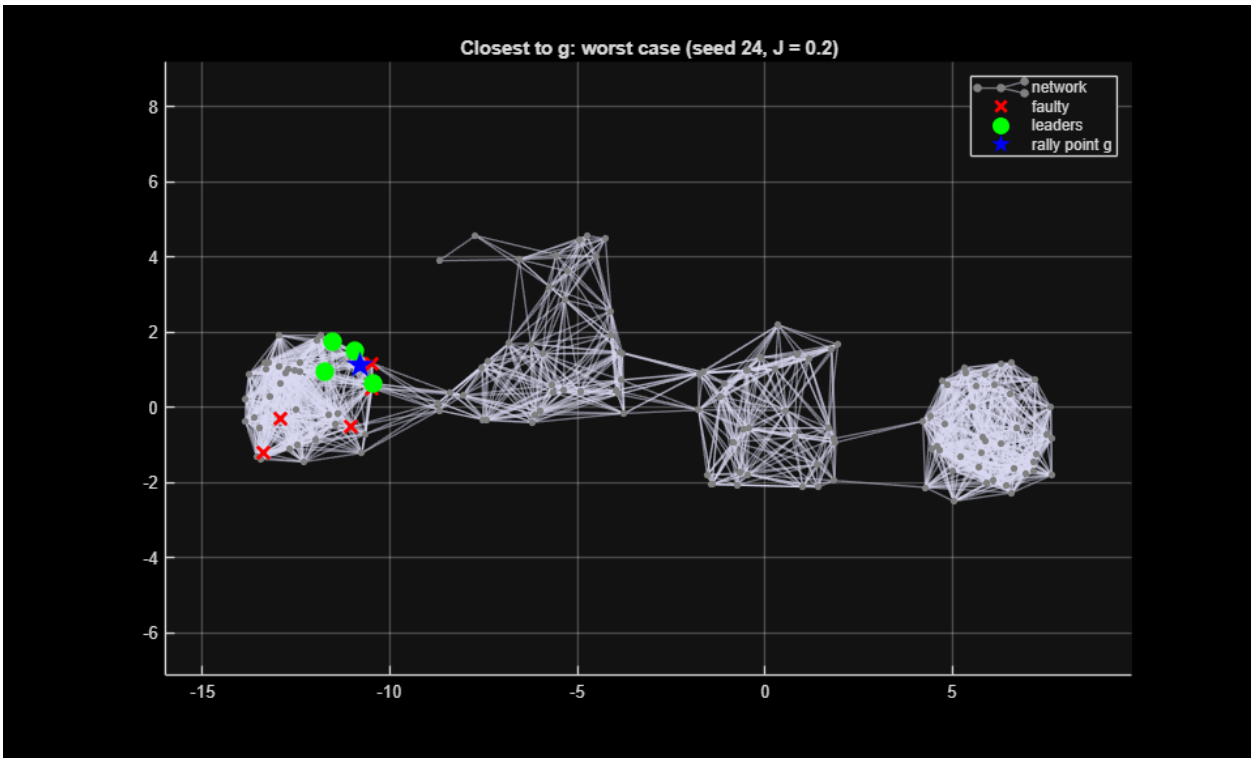
Strategy	Mean	Median	Worst
Random	51.23	52.79	17.69
Highest degree	45.47	46.66	0.92
Closest to g	8.91	5.51	0.19

```
Random worst: seed 1, J = 17.69
Highest degree worst: seed 12, J = 0.92
Closest to g worst: seed 24, J = 0.19
```


Baseline score distributions over 30 practice deployments
Score J (%)
90
80
70
60
50
40
30
20
10
0
Random
Highest degree
Closest to g


Random: worst case (seed 1, J = 17.7)
network
faulty
leaders
rally point g
6
4
2
0
-2
-4
-6
-8
-8
-6
-4
-2
0
2
4
6
8
10
12





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